PROBLEM 12 | Problems T2_CrystStruct.pdf

Crystalline Structures & Miller Indices

Drawing Direction Vectors in Cubic Cells

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12. Draw direction vectors in unit cells for the following cubic directions:

a) $[1\,\bar{1}\,\bar{2}]$
b) $[\bar{3}\,3\,1]$
c) $[2\,\bar{1}\,2]$
d) $[\bar{1}\,0\,1]$
e) $[3\,2\,1]$
f) $[1\,\bar{2}\,\bar{2}]$
g) $[1\,\bar{2}\,3]$
h) $[0\,\bar{2}\,1]$
i) $[2\,\bar{3}\,3]$
j) $[1\,2\,\bar{1}]$
k) $[\bar{2}\,\bar{2}\,3]$
l) $[1\,0\,\bar{3}]$

(No numerical solution is given in the official sheet. The overbars were read from the rendered page of the PDF; in the sheet item k) is printed on a second line after l).)

PHASE 1: HYPOTHESIS & PARAMETERS

Geometric Context:

Twelve crystallographic vectors are drawn inside a standard cubic unit cell of normalised dimensions $1 \times 1 \times 1$.

Quantity	Symbol	Value / unit
Lattice parameter (cube edge)	a	arbitrary, in m; not needed, because every coordinate is a fraction of a
Direction indices	$[u\,v\,w]$	integers, dimensionless
Tail (new origin O')	(x_0, y_0, z_0)	fractions of a, each 0 or 1
Head (arrow tip)	(x_f, y_f, z_f)	fractions of a, each in $[0, 1]$

Hypotheses: cubic cell ($a = b = c, \alpha = \beta = \gamma = 90°$), right-handed axes x, y, z along the cell edges [Session 3 Slide 37]; each direction is drawn as one vector inside one cell, which is always possible because parallel vectors share the same indices [Session 3 Slide 41].

Normalisation and Origin Translation Rules [Session 3 Slides 37-40]:

1. The components of the direction vector must be reduced by dividing by the component of largest absolute value ($u_{\max}$) so that the vector fits strictly inside the volume of a single unit cell:

$$\vec{r} = \left(\frac{u}{u_{\max}}, \frac{v}{u_{\max}}, \frac{w}{u_{\max}} \right)$$

2. If any component is negative (overbar $\bar{u}$), the **origin of the vector must be translated** to the position $+1$ along that axis so that the vector points towards the interior of the cell.

`PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION`

Citing the systematic procedure [Session 3 Slides 39-40]:

1. Identify whether there are negative indices and fix the local origin:
2. $\bar{u} \implies$ origin at $x = 1$.
3. $\bar{v} \implies$ origin at $y = 1$.
4. $\bar{w} \implies$ origin at $z = 1$.
5. Draw the vector from the origin (x_0, y_0, z_0) to the end point $(x_0 + \Delta x, y_0 + \Delta y, z_0 + \Delta z)$.
6. Join both points with an oriented arrow.

`PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP`

For every direction the same four operations are applied [Session 3 Slides 39-40]:

1. $m = \max(|u|, |v|, |w|)$ and reduced vector $\vec{r} = \frac{1}{m}(u, v, w)$, so that the largest component is ± 1;
2. new origin O': a 1 in the position of each negative index, a 0 elsewhere;
3. head $= O' + \vec{r}$, component by component;
4. check: head $-$ tail $= \vec{r}$ and $m\,\vec{r} = (u, v, w)$, and every coordinate of the head lies in $[0, 1]$ (the arrow stays inside the cell).

a) $[1\,\bar{1}\,\bar{2}]$**.** $m = 2, \vec{r} = \frac{1}{2}(1, -1, -2) = (\frac{1}{2}, -\frac{1}{2}, -1)$. Negative v and w, so $O' = (0, 1, 1)$. Head $= (0 + \frac{1}{2},\ 1 - \frac{1}{2},\ 1 - 1) = (\frac{1}{2}, \frac{1}{2}, 0)$, the centre of the bottom face. Check: $2(\frac{1}{2}, -\frac{1}{2}, -1) = (1, -1, -2)$.

b) $[\bar{3}\,3\,1]$**.** $m = 3, \vec{r} = \frac{1}{3}(-3, 3, 1) = (-1, 1, \frac{1}{3})$. Negative u, so $O' = (1, 0, 0)$. Head $= (1 - 1,\ 0 + 1,\ 0 + \frac{1}{3}) = (0, 1, \frac{1}{3})$, one third of the way up the back vertical edge $x = 0, y = 1$. Check: $3(-1, 1, \frac{1}{3}) = (-3, 3, 1)$.

c) $[2\,\bar{1}\,2]$**.** $m=2, \vec{r}=\frac{1}{2}(2,-1,2)=(1,-\frac{1}{2},1)$. Negative v, so $O'=(0,1,0)$. Head $=(0+1,\ 1-\frac{1}{2},\ 0+1)=(1,\frac{1}{2},1)$, the midpoint of the top edge of the front face $x=1$. Check: $2(1,-\frac{1}{2},1)=(2,-1,2)$.

d) $[\bar{1}\,0\,1]$**.** $m=1, \vec{r}=(-1,0,1)$. Negative u, so $O'=(1,0,0)$. Head $=(1-1,\ 0+0,\ 0+1)=(0,0,1)$: the diagonal of the face $y=0$. Check: $m=1$, no scaling.

e) $[3\,2\,1]$**.** $m=3, \vec{r}=\frac{1}{3}(3,2,1)=(1,\frac{2}{3},\frac{1}{3})$. No negative index, so $O'=(0,0,0)$. Head $=(1,\frac{2}{3},\frac{1}{3})$, a point of the front face $x=1$. Check: $3(1,\frac{2}{3},\frac{1}{3})=(3,2,1)$.

f) $[1\,\bar{2}\,\bar{2}]$**.** $m=2, \vec{r}=\frac{1}{2}(1,-2,-2)=(\frac{1}{2},-1,-1)$. Negative v and w, so $O'=(0,1,1)$. Head $=(0+\frac{1}{2},\ 1-1,\ 1-1)=(\frac{1}{2},0,0)$, the midpoint of the x edge. Check: $2(\frac{1}{2},-1,-1)=(1,-2,-2)$.

g) $[1\,\bar{2}\,3]$**.** $m=3, \vec{r}=\frac{1}{3}(1,-2,3)=(\frac{1}{3},-\frac{2}{3},1)$. Negative v, so $O'=(0,1,0)$. Head $=(0+\frac{1}{3},\ 1-\frac{2}{3},\ 0+1)=(\frac{1}{3},\frac{1}{3},1)$, a point of the top face. Check: $3(\frac{1}{3},-\frac{2}{3},1)=(1,-2,3)$.

h) $[0\,\bar{2}\,1]$**.** $m=2, \vec{r}=\frac{1}{2}(0,-2,1)=(0,-1,\frac{1}{2})$. Negative v, so $O'=(0,1,0)$. Head $=(0,\ 1-1,\ 0+\frac{1}{2})=(0,0,\frac{1}{2})$, the midpoint of the z edge; the vector lies in the face $x=0$. Check: $2(0,-1,\frac{1}{2})=(0,-2,1)$.

i) $[2\,\bar{3}\,3]$**.** $m=3, \vec{r}=\frac{1}{3}(2,-3,3)=(\frac{2}{3},-1,1)$. Negative v, so $O'=(0,1,0)$. Head $=(0+\frac{2}{3},\ 1-1,\ 0+1)=(\frac{2}{3},0,1)$, a point of the top edge of the face $y=0$. Check: $3(\frac{2}{3},-1,1)=(2,-3,3)$.

j) $[1\,2\,\bar{1}]$**.** $m=2, \vec{r}=\frac{1}{2}(1,2,-1)=(\frac{1}{2},1,-\frac{1}{2})$. Negative w, so $O'=(0,0,1)$. Head $=(0+\frac{1}{2},\ 0+1,\ 1-\frac{1}{2})=(\frac{1}{2},1,\frac{1}{2})$, the centre of the right face $y=1$. Check: $2(\frac{1}{2},1,-\frac{1}{2})=(1,2,-1)$.

k) $[\bar{2}\,\bar{2}\,3]$**.** $m=3, \vec{r}=\frac{1}{3}(-2,-2,3)=(-\frac{2}{3},-\frac{2}{3},1)$. Negative u and v, so $O'=(1,1,0)$. Head $=(1-\frac{2}{3},\ 1-\frac{2}{3},\ 0+1)=(\frac{1}{3},\frac{1}{3},1)$, a point of the top face. Check: $3(-\frac{2}{3},-\frac{2}{3},1)=(-2,-2,3)$.

l) $[1\,0\,\bar{3}]$**.** $m=3, \vec{r}=\frac{1}{3}(1,0,-3)=(\frac{1}{3},0,-1)$. Negative w, so $O'=(0,0,1)$. Head $=(0+\frac{1}{3},\ 0,\ 1-1)=(\frac{1}{3},0,0)$, a point of the x edge; the vector lies in the face $y=0$. Check: $3(\frac{1}{3},0,-1)=(1,0,-3)$.

Summary

Item	Direction	m	Reduced $\vec{r}$	Tail O'	Head	Where the head lies
a	$[1\,\bar{1}\,\bar{2}]$	2	$(\frac{1}{2},-\frac{1}{2},-1)$	$(0,1,1)$	$(\frac{1}{2},\frac{1}{2},0)$	centre of the bottom face
b	$[\bar{3}\,3\,1]$	3	$(-1,1,\frac{1}{3})$	$(1,0,0)$	$(0,1,\frac{1}{3})$	edge $x=0$, $y=1$
c	$[2\,\bar{1}\,2]$	2	$(1,-\frac{1}{2},1)$	$(0,1,0)$	$(1,\frac{1}{2},1)$	midpoint of edge $x=1$, $z=1$
d	$[\bar{1}\,0\,1]$	1	$(-1,0,1)$	$(1,0,0)$	$(0,0,1)$	corner (face diagonal of $y=0$)
e	$[3\,2\,1]$	3	$(1,\frac{2}{3},\frac{1}{3})$	$(0,0,0)$	$(1,\frac{2}{3},\frac{1}{3})$	face $x=1$
f	$[1\,\bar{2}\,\bar{2}]$	2	$(\frac{1}{2},-1,-1)$	$(0,1,1)$	$(\frac{1}{2},0,0)$	midpoint of the x edge
g	$[1\,\bar{2}\,3]$	3	$(\frac{1}{3},-\frac{2}{3},1)$	$(0,1,0)$	$(\frac{1}{3},\frac{1}{3},1)$	top face $z=1$
h	$[0\,\bar{2}\,1]$	2	$(0,-1,\frac{1}{2})$	$(0,1,0)$	$(0,0,\frac{1}{2})$	midpoint of the z edge
i	$[2\,\bar{3}\,3]$	3	$(\frac{2}{3},-1,1)$	$(0,1,0)$	$(\frac{2}{3},0,1)$	edge $y=0$, $z=1$
j	$[1\,2\,\bar{1}]$	2	$(\frac{1}{2},1,-\frac{1}{2})$	$(0,0,1)$	$(\frac{1}{2},1,\frac{1}{2})$	centre of the face $y=1$
k	$[\bar{2}\,\bar{2}\,3]$	3	$(-\frac{2}{3},-\frac{2}{3},1)$	$(1,1,0)$	$(\frac{1}{3},\frac{1}{3},1)$	top face $z=1$
l	$[1\,0\,\bar{3}]$	3	$(\frac{1}{3},0,-1)$	$(0,0,1)$	$(\frac{1}{3},0,0)$	x edge

Drawings

Each panel shows the cell with the axes x (towards the reader), y (right) and z (up); the filled dot is the tail O', the open dot the head, and a dotted line drops from a head that is not on the base or on a vertical edge to its foot on the plane $z = 0$.

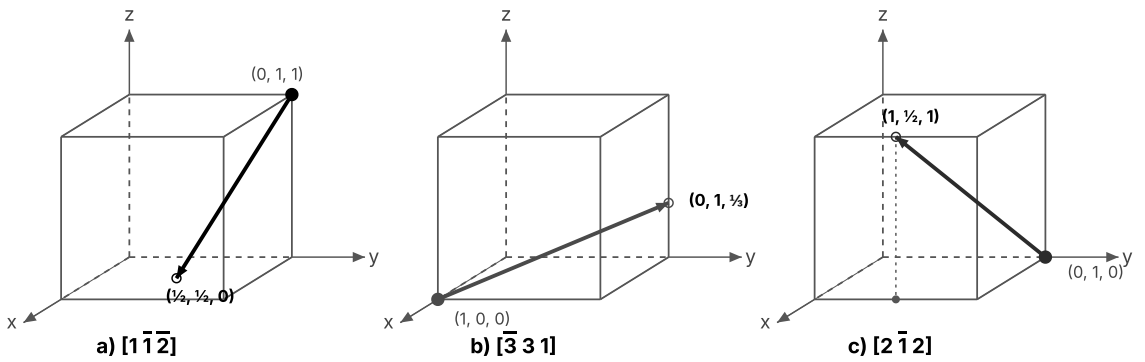


Figure 12.1. Directions a) to c).

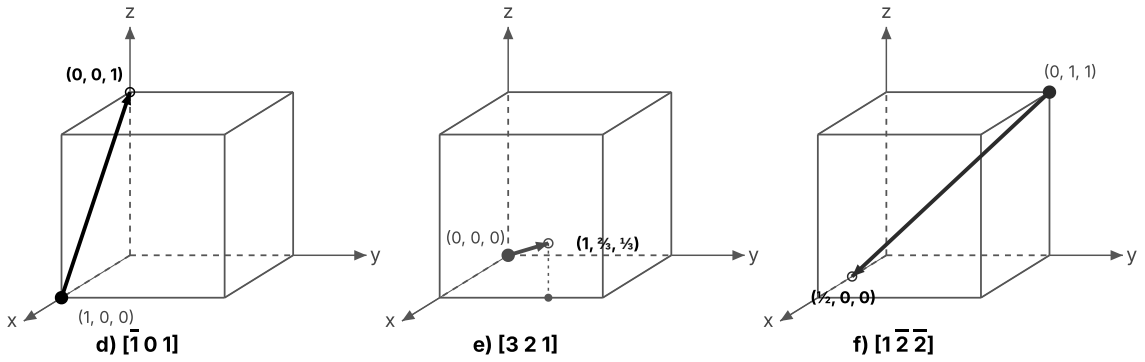


Figure 12.2. Directions d) to f).

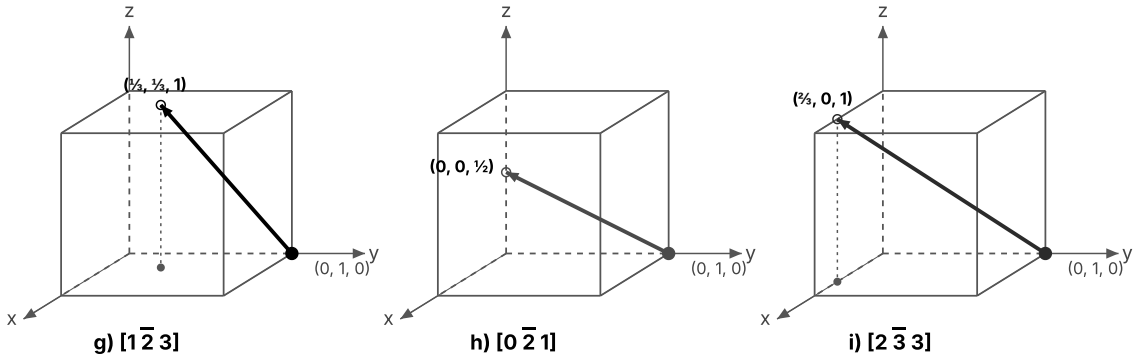


Figure 12.3. Directions g) to i).

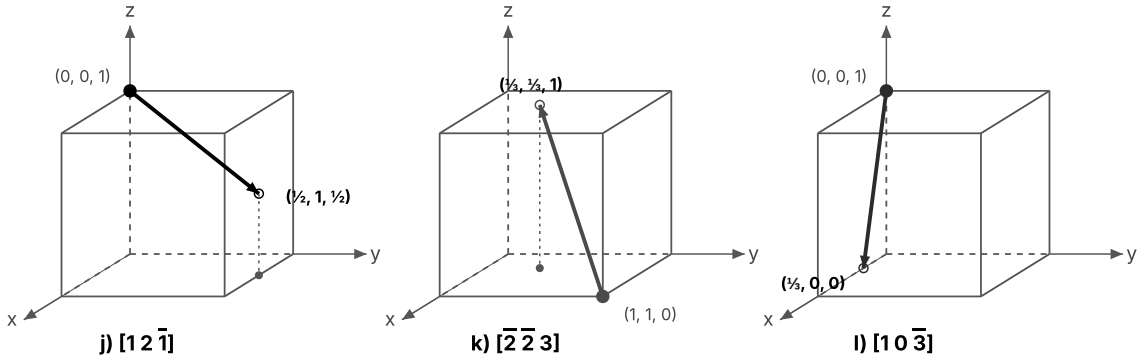


Figure 12.4. Directions j) to l).

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

1. **Dimensions:** the indices are pure numbers and every coordinate is a fraction of the lattice parameter, $x = X/a$ with $[X] = \mathrm{m}$ and $[a] = \mathrm{m}$, so $[x] = \mathrm{m}/\mathrm{m} = 1$. The drawings are therefore valid for any cubic metal; only the real length of each vector scales with a.
2. **Containment check:** all twelve heads have coordinates in $[0, 1]$ (table above), so every arrow stays inside one cell, as the construction requires [Session 3 Slide 39].
3. **Parallelism:** multiplying a vector by a positive scalar does not change the direction: $[1\,\bar{1}\,\bar{2}]$ and $[2\,\bar{2}\,\bar{4}]$ are the **same crystallographic direction** [Session 3 Slide 41]. Reversing the arrow changes every sign: the opposite of $[1\,2\,\bar{1}]$ is $[\bar{1}\,\bar{2}\,1]$.
4. **Families:** in a cubic cell, directions whose indices are permutations of the same numbers, with any signs, belong to the same family. Here $[2\,\bar{1}\,2]$ and $[1\,\bar{2}\,\bar{2}]$ are both members of $\langle 1\,2\,2\rangle$, and $[2\,\bar{3}\,3]$ belongs to $\langle 2\,3\,3\rangle$; members of one family have the same repeat length along the line.
5. **Limiting case:** when $m = 1$ (item d) no scaling is needed and the head is a lattice corner; for $m > 1$ at least one component is a fraction, so the head falls on an edge, a face or (in general) inside the cell.